



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

CIVIL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Civil Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers and small rounding differences.

QUESTION 1

1 1.1 A ✓✓ 1.2 A ✓✓ 1.3 B ✓✓ 1.4 A ✓✓ 1.5 A ✓✓ [10]

QUESTION 2

- 2.1** Any THREE: inspected by a competent person before use (daily) ✓; base plates on firm, level ground ✓; guardrails and toe boards on working platforms ✓; braced and tied to the building; never overloaded; safe ladder access (3)
- 2.2** (a) To set out and transfer levels (heights) on a building site ✓✓, e.g. for foundations and floors. (4)
(b) To compact fresh concrete ✓ by shaking out trapped air bubbles, giving dense, strong concrete ✓.
- 2.3** A person who has the knowledge, training and experience for the specific work ✓ and who knows the Act and its regulations that apply to it ✓. (2)
- 2.4** Any THREE: shore (support) the sides or cut them back to a safe slope ✓; keep spoil and machines at least 1 m from the edge ✓; provide a ladder for getting in and out ✓; barricade the trench; inspect it daily and after rain (3)
- 2.5** A disc rated lower than the grinder's speed spins faster than it was designed for ✓; it can shatter (burst) ✓ and the flying pieces can cause serious injury ✓. (3)

[15]

QUESTION 3

- 3.1.1** Wet volume = $5 \times 4 \times 0,15$ ✓ = 3 m^3 ✓ (3)
Dry volume = $3 \times 1,5$ = $4,5 \text{ m}^3$ ✓
- 3.1.2** Total parts = $1 + 2 + 4$ = 7 (3)
Cement = $4,5 \times 1/7$ = $0,64 \text{ m}^3$ ✓ Sand = $4,5 \times 2/7$ = $1,29 \text{ m}^3$ ✓ Stone = $4,5 \times 4/7$ = $2,57 \text{ m}^3$ ✓
- 3.1.3** Mass = $0,643 \times 1\,440$ = $925,71 \text{ kg}$ ✓ (3)
Bags = $925,71 \div 50$ = $18,51$ ✓ → 19 bags ✓ (must round up)
- 3.2** Place the clean, damp cone on a flat, non-absorbent base and stand on the foot rests ✓. Fill it in three equal layers ✓, rodding each layer 25 times with the tamping rod ✓. Strike off the top level and lift the cone slowly and vertically ✓. Measure the drop from the top of the cone to the highest point of the concrete: this is the slump ✓. (5)
- 3.3** Any TWO: lower strength ✓; more shrinkage and cracking ✓; porous, less durable concrete; segregation and bleeding (2)

- 3.4** Any TWO with purpose: accelerator ✓ – speeds up setting and early strength (cold weather) ✓; retarder ✓ – slows setting (hot weather, long transport) ✓; plasticiser – improves workability without extra water; air-entraining agent – resists frost damage; waterproofing agent – reduces water penetration (4)

[20]

QUESTION 4

- 4.1** Moments about A (clockwise = anticlockwise): (6)
 $R_B \times 6 \checkmark = (10 \times 1,5) + (15 \times 3) + (20 \times 4,5) \checkmark = 150 \text{ kN}\cdot\text{m} \checkmark$
 $R_B = 25 \text{ kN} \checkmark$
 Upward forces = downward forces: $R_A = 45 - 25 \checkmark = 20 \text{ kN} \checkmark$
- 4.2** Flange: $A = 120 \times 20 = 2\,400 \text{ mm}^2 \checkmark$, $y = 100 + 10 = 110 \text{ mm} \checkmark$ (6)
 Web: $A = 20 \times 100 = 2\,000 \text{ mm}^2 \checkmark$, $y = 50 \text{ mm} \checkmark$
 $\bar{y} = (2\,400 \times 110 + 2\,000 \times 50) \div 4\,400 \checkmark = 364\,000 \div 4\,400 = 82,73 \text{ mm}$
 from the bottom ✓
- 4.3** $A = \pi \times 20^2 \div 4 = 314,16 \text{ mm}^2 \checkmark$ (7)
 (a) $\sigma = F \div A = 50\,000 \div 314,16 \checkmark = 159,15 \text{ MPa} \checkmark$
 (b) $\varepsilon = \sigma \div E = 159,15 \div 200\,000 \checkmark = 7,96 \times 10^{-4} (0,000\,796) \checkmark$
 (c) $\Delta L = \varepsilon \times L = 7,96 \times 10^{-4} \times 2\,000 \checkmark = 1,59 \text{ mm} \checkmark$
- 4.4** (a) Stress is the internal resisting force per unit area ✓, $\sigma = F \div A \checkmark$. (6)
 (b) Strain is the change in length divided by the original length ✓, $\varepsilon = \Delta L \div L$ (no units) ✓.
 (c) Within the elastic limit ✓, stress is directly proportional to strain ✓.

[25]

QUESTION 5

- 5.1** Any FOUR: strong and rigid enough to carry the wet concrete without bulging ✓; (4)
 watertight joints so cement paste does not leak ✓; easy to erect and strip ✓; can be re-used ✓; accurate in shape, size and level; smooth inside face; economical
- 5.2** The side formwork carries no load and can be removed after a day or two ✓. The soffit (2)
 carries the full weight of the beam until the concrete has gained enough strength (about 14 to 21 days) ✓.
- 5.3** Any FOUR: tie beam ✓; principal rafters ✓; king post ✓; struts ✓; ridge; purlins; wall (4)
 plate
- 5.4** A strip foundation is a continuous strip of concrete under the load-bearing walls ✓. A raft (4)
 foundation is a reinforced concrete slab under the whole building ✓. A raft is used where the soil has a low bearing capacity ✓ or where uneven settlement is expected, e.g. clay or made-up ground ✓.
- 5.5** Any TWO: stops rain penetrating to the inside wall ✓; better heat and sound insulation (3)
 ✓; less condensation. Wall ties tie the two leaves together so the wall is stable ✓.

- 5.6 (a) $2\,700 \div 180 = 15$ risers ✓ (3)
(b) $2 \times 180 + G = 600$ ✓ $\rightarrow G = 240$ mm ✓

[20]

QUESTION 6

- 6.1 Floor area = $4,2 \times 3,6 = 15,12$ m² ✓; tile area = $0,4 \times 0,4 = 0,16$ m² ✓ (5)
 $15,12 \div 0,16 = 94,5 \rightarrow 95$ tiles ✓
 $95 \times 1,1 = 104,5$ ✓ $\rightarrow 105$ tiles ✓ (accept 104 if 10% was added before rounding)
- 6.2 Any FIVE: title of the drawing ✓; name of the client or project ✓; scale ✓; drawing number ✓; date ✓; name of the draughtsperson; site address; checked / approved by; sheet number (5)

[10]

TOTAL: 100

END OF MARKING GUIDELINE